

CHARLWOOD, United Kingdom

WMO: 037690

Lat: 51.144N

Lon: 0.229W

Elev: 68

StdP: 100.51

Time Zone: 0.00 (EUW)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-4.4	-3.0	-6.3	2.2	-2.3	-5.0	2.5	-1.4	9.8	6.5	9.0	7.0	1.1	10	0.528

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	27.8	19.7	25.6	18.5	23.6	17.6	20.4	26.5	19.2	24.2	18.3	22.3	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.2	13.2	22.5	17.4	12.5	21.1	16.6	11.9	20.1	58.8	26.6	54.8	24.2	51.8	22.4	25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.3	7.3	6.4	DB	-7.7	31.3	1.9	2.1	-9.1	32.8	-10.3	34.1	-11.4	35.3	-12.8	36.8	(4)
(5)			WB	-8.0	22.5	1.9	1.3	-9.4	23.4	-10.6	24.2	-11.6	24.9	-13.1	25.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	10.4	4.6	4.7	6.5	9.3	12.1	15.3	17.4	16.8	14.3	11.2	7.4	5.2	(6)
(7)		DBStd	5.45	3.37	3.20	3.02	2.98	2.88	2.54	2.48	2.35	2.50	3.02	3.32	3.69	(7)
(8)		HDD10.0	753	168	151	112	46	11	0	0	0	1	22	89	153	(8)
(9)		HDD18.3	2924	425	382	367	271	194	95	49	61	122	222	328	407	(9)
(10)		CDD10.0	912	1	2	4	25	76	161	229	209	131	59	11	4	(10)
(11)		CDD18.3	40	0	0	0	0	1	6	20	12	2	0	0	0	(11)
(12)		CDH23.3	491	0	0	0	4	17	86	224	131	27	3	0	0	(12)
(13)		CDH26.7	113	0	0	0	0	1	17	58	35	2	1	0	0	(13)
(14)	Wind	WSAvg	2.8	3.3	3.2	3.3	2.9	2.8	2.7	2.6	2.4	2.4	2.7	2.7	3.0	(14)
(15)	Precipitation	PrecAvg	762	80	51	57	54	52	57	48	57	69	77	82	81	(15)
(16)		PrecMax	1058	183	143	118	113	142	137	103	131	224	196	196	146	(16)
(17)		PrecMin	508	14	7	4	4	1	7	10	1	12	3	15	15	(17)
(18)		PrecStd	104	44	33	27	30	29	35	26	32	50	49	45	37	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.4	14.6	18.3	23.7	25.3	28.9	31.2	30.5	26.3	20.8	15.9	13.6	(19)
(20)			MCWB	11.1	10.5	11.8	15.6	17.7	20.8	21.3	21.5	19.0	16.8	14.5	12.2	(20)
(21)		2%	DB	11.6	11.9	15.1	19.9	22.9	25.6	28.2	26.8	23.4	18.3	14.6	12.6	(21)
(22)			MCWB	10.6	9.7	10.7	13.8	16.6	18.7	19.6	19.7	17.8	15.4	13.2	11.5	(22)
(23)		5%	DB	10.7	10.7	13.2	17.3	20.5	23.4	25.9	24.1	21.2	17.0	13.6	11.8	(23)
(24)			MCWB	9.7	9.0	9.7	12.3	15.2	17.4	18.6	18.2	16.6	14.7	12.2	10.7	(24)
(25)		10%	DB	9.9	9.7	11.5	15.3	18.3	21.5	23.5	22.0	19.5	15.9	12.7	10.9	(25)
(26)			MCWB	8.8	8.2	9.0	11.1	13.7	16.5	17.6	17.0	15.6	14.1	11.4	9.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	11.7	11.3	12.4	16.2	19.0	21.4	22.1	22.0	20.0	17.4	14.9	12.6	(27)
(28)			MCDWB	12.0	12.9	15.7	22.5	23.6	28.1	29.8	29.4	24.7	19.7	15.5	13.2	(28)
(29)		2%	WB	10.9	10.3	11.3	14.2	17.1	19.4	20.2	20.1	18.6	16.1	13.5	11.8	(29)
(30)			MCDWB	11.5	11.4	14.0	19.0	21.6	24.3	26.8	25.8	22.1	17.5	14.3	12.4	(30)
(31)		5%	WB	9.9	9.3	10.5	12.8	15.6	18.1	19.2	18.7	17.4	15.3	12.6	11.0	(31)
(32)			MCDWB	10.5	10.4	12.7	16.6	19.7	22.4	24.7	22.6	20.0	16.5	13.4	11.7	(32)
(33)		10%	WB	9.0	8.5	9.5	11.6	14.3	16.9	18.1	17.7	16.4	14.5	11.6	9.9	(33)
(34)			MCDWB	9.8	9.6	11.4	14.5	17.5	20.7	22.6	20.8	18.6	15.6	12.5	10.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.5	8.2	10.1	10.1	10.4	10.6	10.1	10.0	8.1	6.9	6.1	(35)
(36)			MCDBR	6.0	8.1	11.4	14.2	13.8	14.5	15.2	13.8	13.1	8.8	6.9	5.2	(36)
(37)			MCWBR	5.8	6.0	7.5	8.7	8.1	8.3	7.6	7.2	7.8	6.3	5.9	5.0	(37)
(38)		5% WB	MCDBR	5.8	6.9	10.0	12.8	12.4	13.2	13.4	12.2	10.8	7.5	6.7	5.2	(38)
(39)			MCWBR	5.8	5.7	7.1	8.0	7.6	8.1	7.2	6.9	6.9	5.8	6.2	5.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.329	0.345	0.382	0.401	0.405	0.405	0.400	0.393	0.376	0.359	0.333	0.319	(40)	
(41)		taud	2.372	2.377	2.269	2.240	2.272	2.297	2.317	2.358	2.398	2.423	2.417	2.362	(41)	
(42)		Ebn at Noon	682	777	808	837	850	851	849	837	810	749	674	629	(42)	
(43)		Edh at Noon	64	82	109	125	127	125	120	110	95	77	61	55	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.82	1.50	2.65	4.18	5.03	5.57	5.29	4.41	3.29	1.88	0.98	0.64	(44)	
(45)		RadStd	0.07	0.20	0.34	0.42	0.32	0.52	0.46	0.34	0.26	0.17	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page